

Homework (Jalapeno)

- Q1.** The ancient Egyptians built a pyramid with a height h and a base b . If the volume V of the pyramid is given by $V = \frac{1}{3}bh^2$ and the base is 10 times the height, what is the equation for V in terms of h ?
- (A) $V = \frac{10}{3}h^3$
(B) $V = \frac{1}{3}h^2$
(C) $V = 10h^2$
(D) $V = \frac{1}{10}h^3$
(E) $V = h^3$
- Q2.** Solve the system: $x + y = 5$ and $x^2 + y^2 = 25$. What is x ?
- (A) 0
(B) 1
(C) 2
(D) 3
(E) 4
- Q3.** In ancient Greece, a sculptor created a statue with a quadratic equation $x^2 + 4x + 4 = y$. If $x + 2 = 0$, what is y ?
- (A) 0
(B) 4
(C) 8
(D) 12
(E) 16
- Q4.** The equation $x^2 - 4x - 5 = 0$ represents the number of ancient artifacts found in a tomb. If $y = x + 1$, what is y when $x = -1$?
- (A) 0
(B) 1
(C) 2
(D) 3
(E) 4
- Q5.** The quadratic equation $y = x^2 - 3x - 2$ represents the trajectory of a projectile in ancient warfare. If $y = -2$, what is x ?
- (A) -1
(B) 1
(C) 2
(D) 3
(E) 4
- Q6.** Solve the system: $x^2 + 2x + 1 = y$ and $y = x + 1$. What is the intersection point?
- (A) $(-1, 0)$
(B) $(-2, -1)$
(C) $(0, 1)$
(D) $(1, 2)$
(E) $(2, 3)$
- Q7.** The ancient civilization of Babylon used the quadratic equation $x^2 + 2x - 6 = 0$ to represent the growth of their population. If x represents the number of years, what is x ?
- (A) -3
(B) -2
(C) 1
(D) 2
(E) 3
- Q8.** The equation $y = x^2 + 2x - 3$ represents the design of an ancient bridge. If $y = 0$, what is x ?
- (A) -3
(B) -1
(C) 1
(D) 2
(E) 3

Open Ended Questions (FRQ)

- Q9.** The ancient Chinese used the quadratic equation $x^2 - 4x + 3 = y$ to represent the trajectory of a projectile. If $y = x - 1$, solve the system and find the intersection point. Show all work and explain your answer.
- Q10.** The quadratic equation $y = x^2 + 2x - 6$ represents the growth of a population in ancient Egypt. If x represents the number of years, find the value of x when $y = 0$. Show all work and explain your answer.

Chef's Tips

- Tip 1: Encourage students to use graphs to visualize the intersection points.
- Tip 2: Remind students to check their work by plugging their answers back into the original equations.
- Tip 3: Suggest that students use different methods, such as substitution or elimination, to solve the systems of equations.